Semantic Communication + AI

Learning what to send for a remote image-classification task

An edge camera sees an image. A server across a limited, noisy link must recognize it. We compare a normal communication system with systems that learn what information to send.

Capstone First Review

Review window: 18–22 August 2026

Presentation slot: 15 minutes

Team and guide details to be added

What problem are we solving?

1
Observe
An edge camera sees an image.

2
Communicate
The link has limited bandwidth and added noise.

3
Classify
A server decides what the image contains.

4
Question
Must the camera send every image detail?

Why this matters

Sending a complete high-quality image can be expensive when the receiver only needs a correct task decision.

The project asks whether a learned system can preserve that decision using the same channel budget.

How is this normally done?

Image (camera input) → **Compress** (JPEG 2000) → **Protect bits** (LDPC + modulation) → **Add noise** (simulated channel) → **AI classifier** (recovered image → ResNet-18)

Compression	reduces the number of source bits.
Error correction	adds controlled redundancy so damaged bits can be recovered.
Modulation	maps protected bits to symbols sent through the channel.
AI classifier	a frozen ResNet-18 maps the recovered image to class logits.

Limitation: the communication stack is designed separately from the AI objective, even though task accuracy is the final metric.

What are we proposing?

Image	→	Learned sender	→	Noisy link	→	Learned receiver	→	Class
sender input		decides what to send		limited bandwidth		recovers task information		final decision

The basic idea

Train the sender and receiver together using labeled images and simulated noise. The system can then learn which image information helps the receiver make the correct decision.

Why not send the label?

The project fixes the deployment split: the sender has an encoder, while the receiver owns the decoder and classifier. Sending a label would move the full task model to the sender and test a different system.

Technical name: deep joint source–channel coding (DJSCC)

AIML method: supervised end-to-end training · model and loss details follow in the methodology

Literature survey and research gap

Short-packet communication	Finite packets pay coding overhead and can fail abruptly.	Measure the practical baseline instead of assuming ideal capacity. [1–4]
Learned image compression	Neural transforms learn compact image representations.	Good reconstruction quality is not the same as good AI task accuracy. [11–17]
DeepJSCC	Neural encoders and decoders can learn continuous channel mappings.	Most image studies emphasize reconstruction rather than classification. [5–10,24]
Task-oriented edge AI	Learned features can preserve inference under link limits.	Representation learning and joint coding are often not isolated. [18–23]
Gap addressed	A controlled three-way experiment that separates neural representation learning from joint channel coding at equal channel use.	

Problem statement, research question and objectives

Problem statement	An edge camera has a limited noisy link, while the server only needs a correct class decision. A conventional system tries to rebuild the image first. The project tests whether a learned system can protect the decision more directly.
Research question	At the same bandwidth and channel conditions, can the learned joint system preserve classification accuracy better than a tuned conventional system, and does joint training explain any remaining gain?

Objectives

1. Build and train the learned sender and receiver models.
2. Build a digital feature system that separates task awareness from joint training.
3. Match channel use, SNR, image identity and noise across all systems.
4. Freeze all learned models and choices, then run one paired test campaign.

Completion depends on executing the fixed protocol correctly, not on obtaining a favorable result.

Proposed methodology

Imagenette-160 · same image and split · same complex-symbol budget · same SNR · same keyed noise

System	Transmission path	Scientific role
Classical + AI	JPEG 2000 → LDPC → AWGN → recovered image → frozen ResNet-18	Conventional reference
AI feature control	Shared neural encoder → quantization → same digital PHY → shared task head	Isolates representation learning
Neural DJSCC	Residual encoder → normalization → AWGN → dual-head neural decoder	Tests end-to-end neural coding

Evaluation controls Train with supervised CE + λMSE; tune on validation only; freeze before test access; retain failed transmissions; compare paired per-image AI outcomes.

The test split remains sealed until gate G-12. No test result exists at First Review.

Research hypotheses

Primary hypothesis (H1)

At difficult low-SNR conditions, the learned joint system will preserve classification accuracy better than the tuned conventional system across a consecutive region—not at only one isolated point.

Formal decision rule

Comparator: learned versus classical_adaptive at the headline ratio. A point qualifies when $\sqrt{N}\cdot\Delta(s)/\sigma(s) > 1.96$. R_obs is the longest consecutive qualifying run at or below the training SNR. H1 is supported only if $R_{obs} \geq 3$ and the calibrated run p-value ≤ 0.05.

Effect size: the mean paired accuracy difference over the full low-SNR region. A curve crossing is reported if observed, but is not required.

H2	Does a fixed conventional configuration show a cliff while the learned system degrades more gradually?
H3	Does the learned–classical accuracy gap contract as SNR improves?
H4	Does the learned joint system also exceed the task-aware digital feature control?

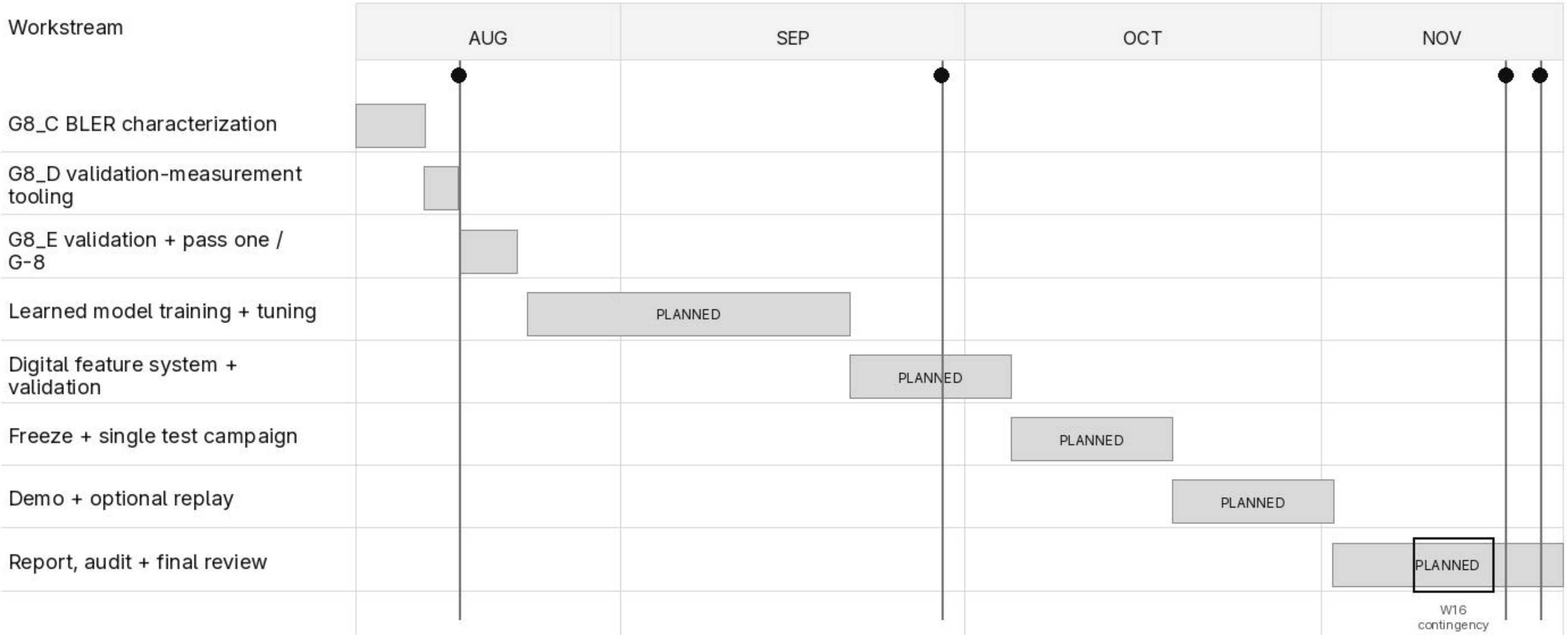
Outcome rule: support, no support or an adverse result are all valid findings. The protocol does not change after test access.

Preliminary work and feasibility

Item	Observed evidence	What it tells us	What it does not tell us
AI task	ResNet-18 from scratch: 89.8% validation	The classification task and clean reference model are viable.	Validation only; test is sealed. (G-1)
Learned model	DJSCC: 1.64 M parameters	The planned model fits the available RTX 4060.	A feasibility profile, not training. (G-7)
Digital link	Golden vectors and BLER conformance pass	The channel-coding implementation behaves as expected.	Conformance evidence, not the full table. (G-2)
Full pipeline	Bounded conventional run completes	Compression, coding, channel and records work together.	Integration evidence, not a comparison. (W4)
G8_C	BLER characterization complete	Pascal successor table is frozen from measured points.	No learned-system result. (G8_C)
G8_D	Validation tooling complete	D0–D7 GREEN; codec/cache/record/resume tooling is ready.	Bounded smoke is not the full campaign. (G8_D)

G8_E/E0 is released but full validation/pass one has not started. Training/fine-tuning, learned-vs-classical results and test evaluation are not available.

Project Gantt chart



Review dates: 18–22 Aug · 29 Sep–3 Oct · 17–21 Nov · report due 20 Nov

G8_C and G8_D are complete. G8_E/E0 is next but not started; all later bars remain planned. W16 is report contingency.

Applications, project scope and risks

Edge AI application

Remote cameras and edge sensors send task-relevant information to a receiver-side AI model when bandwidth, latency or link reliability is limited. This represents split edge/cloud inference.

AI/ML contribution

Channel-robust neural representation learning; a residual encoder and dual-head decoder; CE + λMSE multi-task training; and a digital feature control that isolates the value of joint coding.

Main risks and controls

BLER work is incomplete: resume from the authenticated checkpoint.

A hypothesis may not be supported: report the result without retuning.

Hardware adds confounders: keep it as gated stretch work.

Technical claim boundary Tier 1 is a simulation-based AI experiment over normalized AWGN. OpenJPEG 2.5.4 and TS 38.212-derived LDPC are controls; no complete 5G NR radio or required hardware is claimed.

Outstanding human item: the guide's dated acknowledgement of the no-required-hardware path remains pending.

Summary and next steps

Project summary

Problem	Edge AI inference must operate over a limited noisy link.
AI/ML method	Train a residual neural encoder and dual-head decoder through a differentiable channel.
Evaluation	Measure Imagenette-160 classification at matched channel use against two controlled baselines.
Next work	Finish characterization, train and freeze the models, then run the paired test campaign once.

Six-criterion rubric map

Motivation	2–4, 6, 11
Objectives	6–7, 11–12
Hypothesis	8
Problem Survey	5
Subject Knowledge	3–5, 7–9, 11
Time Plan	10

First Review position The AI/ML method, controlled evaluation protocol, feasibility evidence and completion plan are defined. Final neural training and learned-versus-classical results remain future work.